

internal-external rotation 0 degree-abducted — in vivo – ex vivo difference, 95% CI (overlap only)

Sternoclavicular — Protraction(+)/retraction(-)

single group

Sternoclavicular — Depression(+)/elevation(-)

single group

Sternoclavicular — Backwards(+)/forward(-) rot

single group

Acromioclavicular — Protraction(+)/retraction(-)

single group

Acromioclavicular — Medial(+)/lateral(-) rot

single group

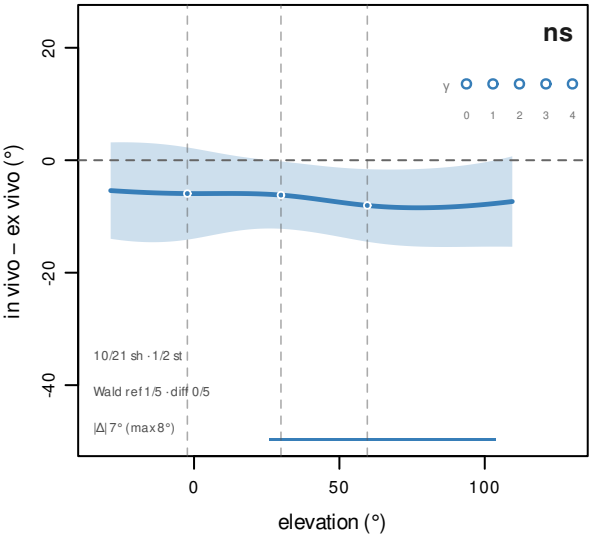
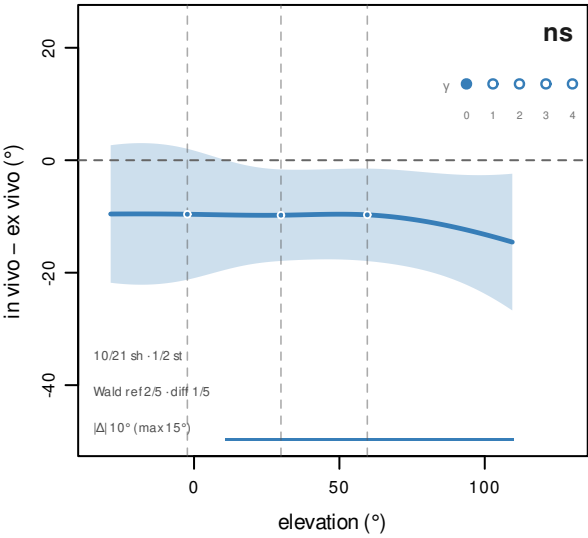
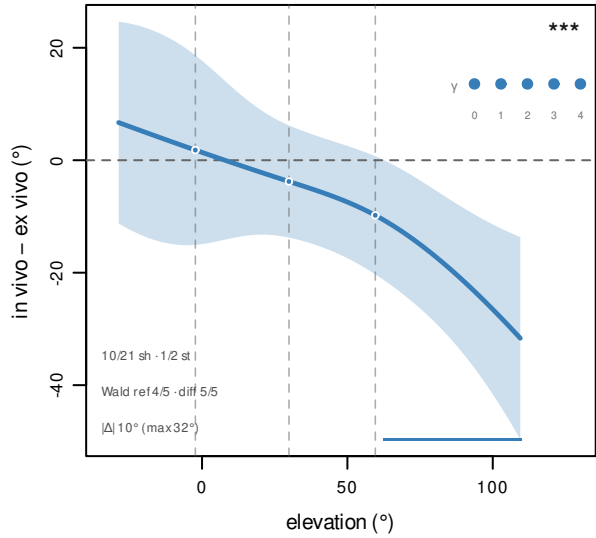
Acromioclavicular — Posterior(+)/anterior(-) tilt

single group

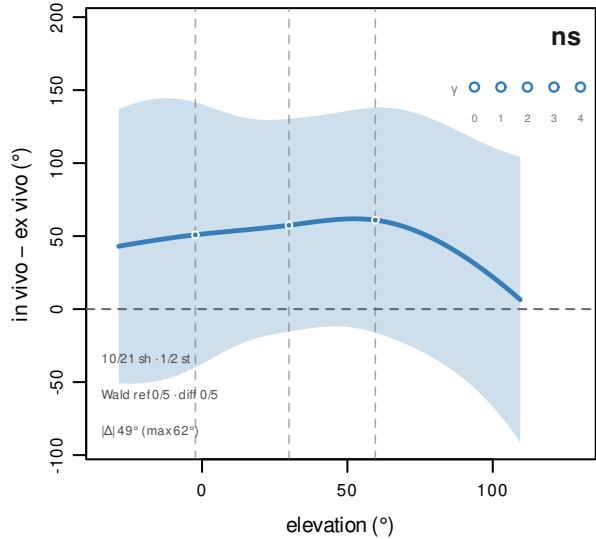
Scapulothoracic — Protraction(+)/retraction(-)

Scapulothoracic — Medial(+)/lateral(-) rot

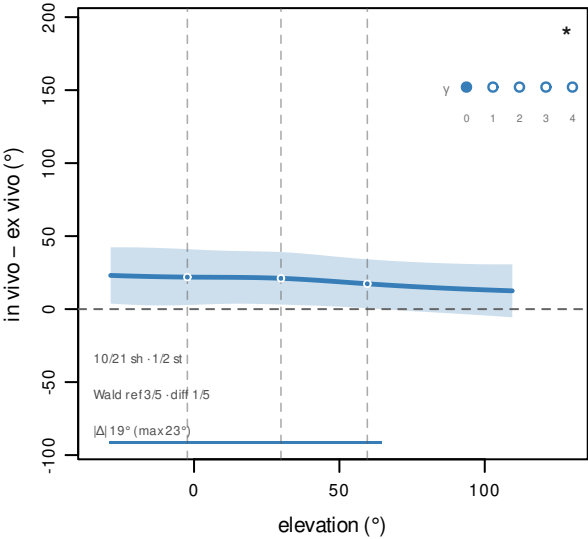
Scapulothoracic — Posterior(+)/anterior(-) tilt



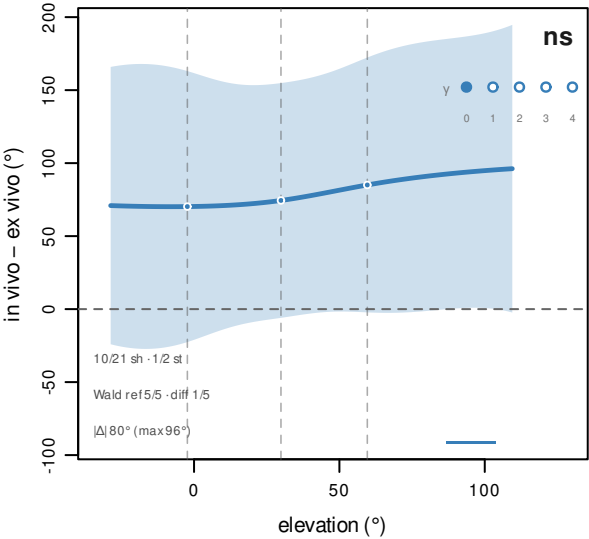
Glenohumeral — Plane of elevation



Glenohumeral — Elevation(-)/Depression(+)



Glenohumeral — Int(+)/ext(-) rotation



blue line = in vivo – ex vivo, band = 95% CI · squares along the bottom = band excludes 0 · dashed verticals = interior knots, dots = value at each knot

top-right: one dot per in-vivo difference coefficient, read LEFT TO RIGHT as gamma_0, gamma_1, ... — the index is printed under each dot. gamma_0 = constant level shift (condin vivo); gamma_k = ns_k:condin vivo, a shape term

filled = that coefficient passed |coef/se| >= 1.96, hollow = did not. stars = JOINT test on ALL difference coefficients at once, BH-FDR: *** p<0.001, ** p<0.01, * p<0.05, ns = not significant

bottom-left: shoulders ex/in · studies ex/in | Wald ref/diff counts | mean and max |Δ| in degrees. Curves are drawn only where ≥2 shoulders have data.